

Open Project: Phase 1

Project Proposal & Exploratory Data Analysis (EDA)

CISC 484/684: Introduction to Machine Learning

Proposal Due: March 15
Weekly Updates Start: March 29

1 Overview

The project proposal is the foundation of your semester work. Its purpose is to formalize your research question, verify your dataset's scale, and establish a rigorous benchmarking framework. For Senior/Graduate level credit, projects must move beyond simple “out-of-the-box” implementation and focus on **model comparison, error analysis, and technical justification**.

2 Proposal Requirements (10 Points)

Your proposal must be a 2–3 page document covering the following four pillars:

2.1 1. Problem Definition & Scientific Hypothesis

- **Task Identification:** Clearly categorize the task (e.g., Multi-task Learning, Out-of-Distribution Detection, or Generative Modeling).
- **The Research Question:** State a testable hypothesis. *Example: “We hypothesize that a Transformer-based architecture will outperform a baseline LSTM in capturing long-range dependencies in this financial time-series dataset.”*

2.2 2. Data Audit & Initial EDA

- **Scale Verification:** Confirm data meets core requirements ($\geq 10,000$ instances OR high-dimensionality).
- **EDA Visualization:** Provide at least two plots (e.g., class distribution, feature correlation heatmap, or PCA projection). You must prove you have successfully loaded and parsed the data.
- **Preprocessing:** Outline your strategy for handling missing values, scaling, or class imbalance.

2.3 3. Comparative Methodology

Identify the models you will use to quantify the “performance premium” of complexity:

- **Phase 1 Baseline:** A classical algorithm (e.g., Random Forest, XGBoost, or SVM or CNN) to establish the performance floor.

- **Phase 2 Advanced Candidate:** A sophisticated architecture (e.g., ResNet, ViT, Diffusion Model) you intend to implement and optimize.

2.4 4. Evaluation & Success Metrics

- Define metrics relevant to your domain (e.g., F1-Score, ROC-AUC, or Inference Latency).
- Describe your validation strategy (e.g., 5-fold cross-validation or stratified temporal splits).

3 Execution Phase: Weekly Progress Reports (10 Points)

Starting the week **after** the proposal submission, each team must submit a status update to Canvas. These ensure the month-long project stays on schedule.

- **Timeline:** Due every **Sunday by 11:59 PM**.
- **Late Policy:** Late submissions **will not be counted** toward the 10% update grade.

Each update must follow the **Status-Obstacle-Plan** format:

1. **Progress Made:** Summarize concrete tasks (e.g., “Trained XGBoost baseline, achieved 82% accuracy”).
2. **Obstacles:** Identify technical hurdles (e.g., “Encountering vanishing gradients,” “Data loading is bottlenecking training”).
3. **Next Steps:** Define milestones for the coming 7 days.

4 Submission & Approval

- **Format:** Single PDF via Canvas. One submission per team.
- **Approval Status:**
 - **Approved:** Proceed to Phase 2.
 - **Provisionally Approved:** Address minor technical feedback in your first weekly update.
 - **Rejected:** The project scope or data is insufficient. A revised proposal is due within 72 hours.